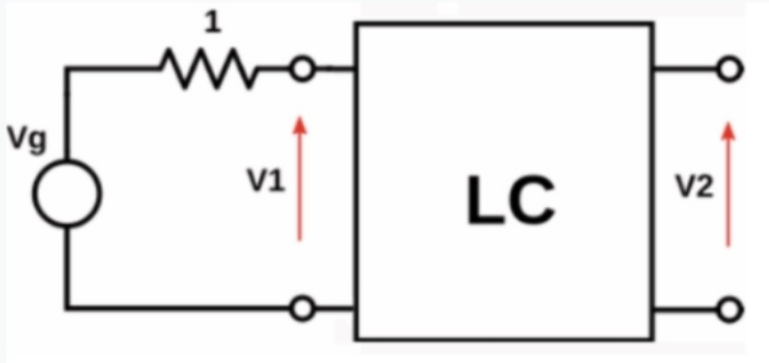


1) Dada la siguiente **transfereencia de tensiones**:



$$T(s) = \frac{V_2}{V_g} = \frac{k \cdot s \cdot (s^2 + 1/16)}{s^3 + 2s^2 + 2s + 1}$$

a) Sintetizar el cuadripolo pasivo sin pérdidas.

b) Hallar el valor de k que cumple con la síntesis y valor de los componentes hallados.

$$Z = \begin{cases} V_1 = z_{11}I_1 + z_{12}I_2 \\ V_2 = z_{21}I_1 + z_{22}I_2 \end{cases}$$

$\boxed{I_2 = 0}$
condición de carga

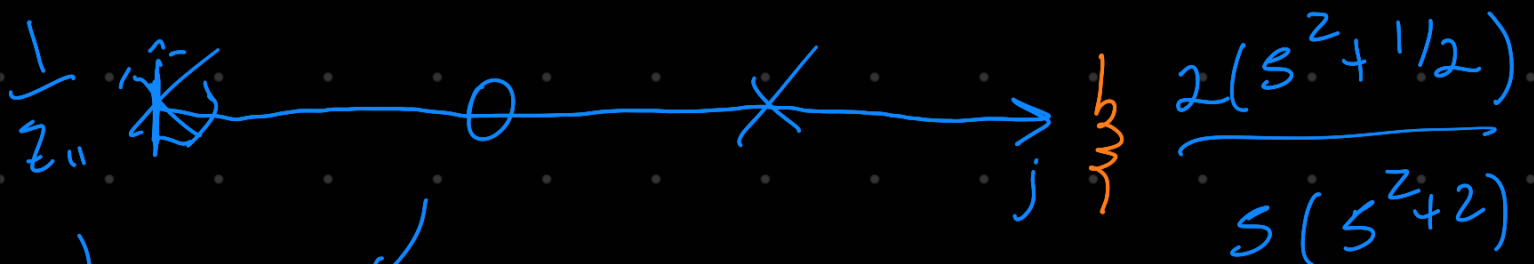
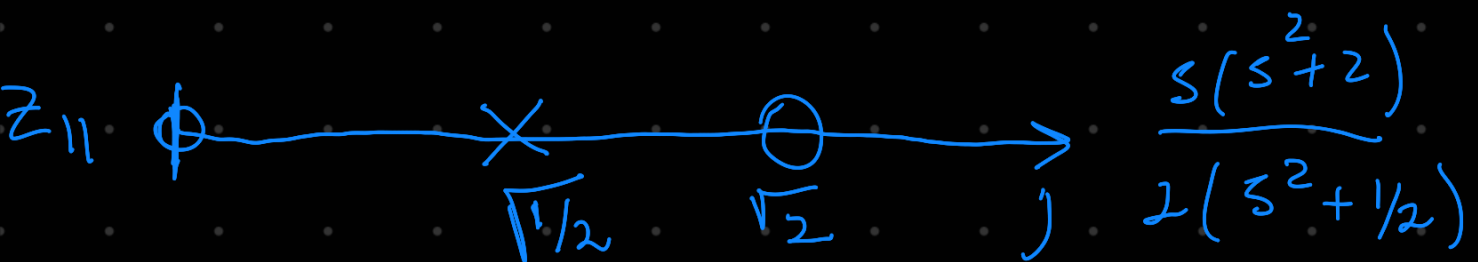
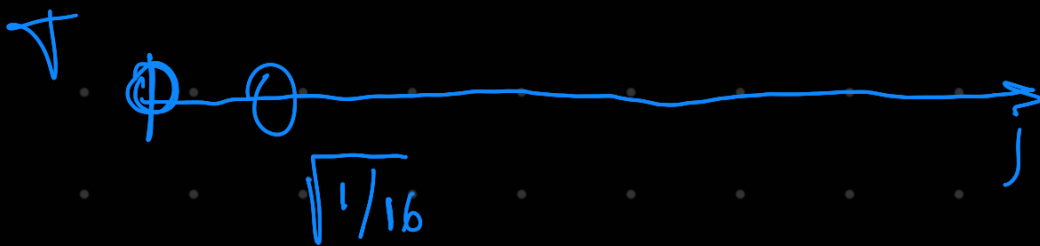
$$V_1 = V_g \frac{z_{11}}{z_{11} + 1}$$

$$\frac{V_2}{V_1} = \frac{z_{21}}{z_{11}}$$

$$\frac{V_2}{V_g} = \frac{\cancel{z_{11}}}{(z_{11} + 1)} \frac{z_{21}}{\cancel{z_{11}}} = k \frac{s(s^2 + 1/16)}{s^3 + 2s^2 + 2s + 1}$$

$$\frac{V_2}{V_0} = \frac{Z_{21}}{Z_{11} + 1} = K \frac{s^3 + s^{1/16}}{s^3 + 2s^2 + 2s + 1} = \frac{P}{Q} \quad \text{Impar}$$

$$\frac{P}{Q} = \frac{P}{\underbrace{N_Q}_{\text{Par}} + \underbrace{N_Q}_{\text{Impar}}} = \frac{s^3 + s^{1/16}}{(2s^2 + 1) + (s^3 + 2s)} = \frac{s^3 + s^{1/16}}{2s^2 + 1 + \underbrace{s^3 + 2s}_{Z_{11}}}$$

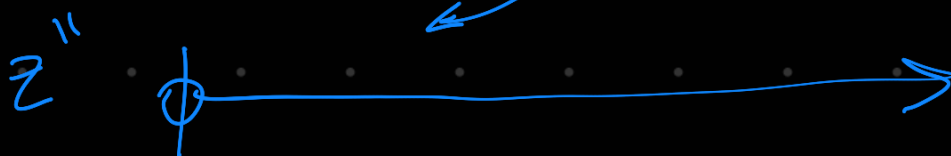
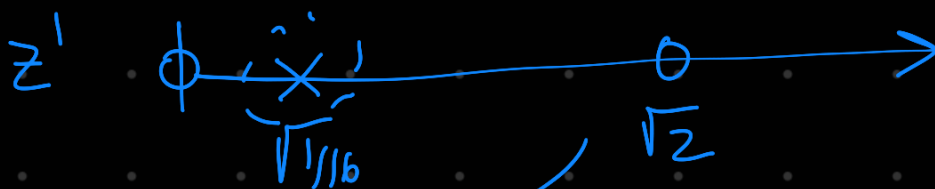
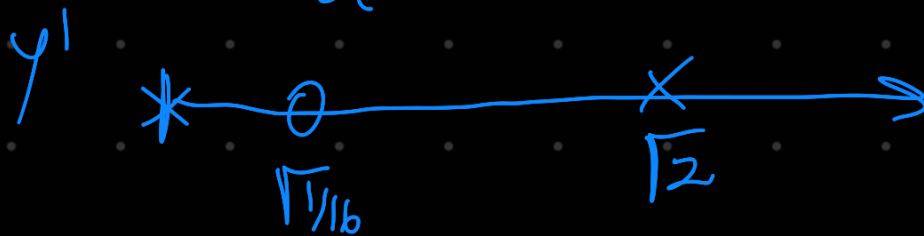


$$Y' = \frac{1}{Z_{11}} - \frac{1}{s} \frac{1}{Z_{11}} \Big|_{s^2 = -\frac{1}{16}}$$

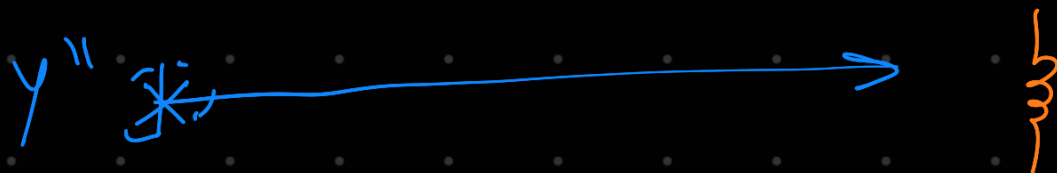
$$Y' = Y_{II} - \frac{1}{5} \frac{14}{31} = \frac{2(s^2 + 1/2) - (s^2 + 2) 14/31}{s(s^2 + 2)}$$

$$2s^2 + 1 - s^2 \frac{14}{31} - \frac{28}{31} = \frac{\frac{48}{31}s^2 + \frac{3}{31}}{s(s^2 + 2)}$$

$$Y' = \frac{\frac{48}{31} \left(s^2 + \frac{3}{48} \cdot 16 \right)}{s(s^2 + 2)} \quad \left. \vphantom{\frac{48}{31}} \right\} \frac{31}{14}$$

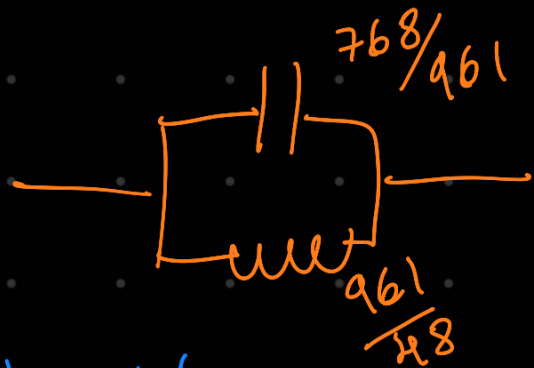


$$\omega^2 = 1/16$$



$$Z'' = \frac{31}{48} \frac{s(s^2 + 2)}{s^2 + 1/16} = \frac{s \frac{961}{768}}{(s^2 + 1/16)}$$

$$V_{1/16} = Z' \frac{s^2 + 1/16}{s} \Big|_{s^2 = -1/16} = \frac{31}{48} \left(-\frac{1}{16} + 2 \right) = \frac{961}{768}$$



$$Z = \left(sC + \frac{1}{sL} \right)^{-1}$$

$$LC = 16$$

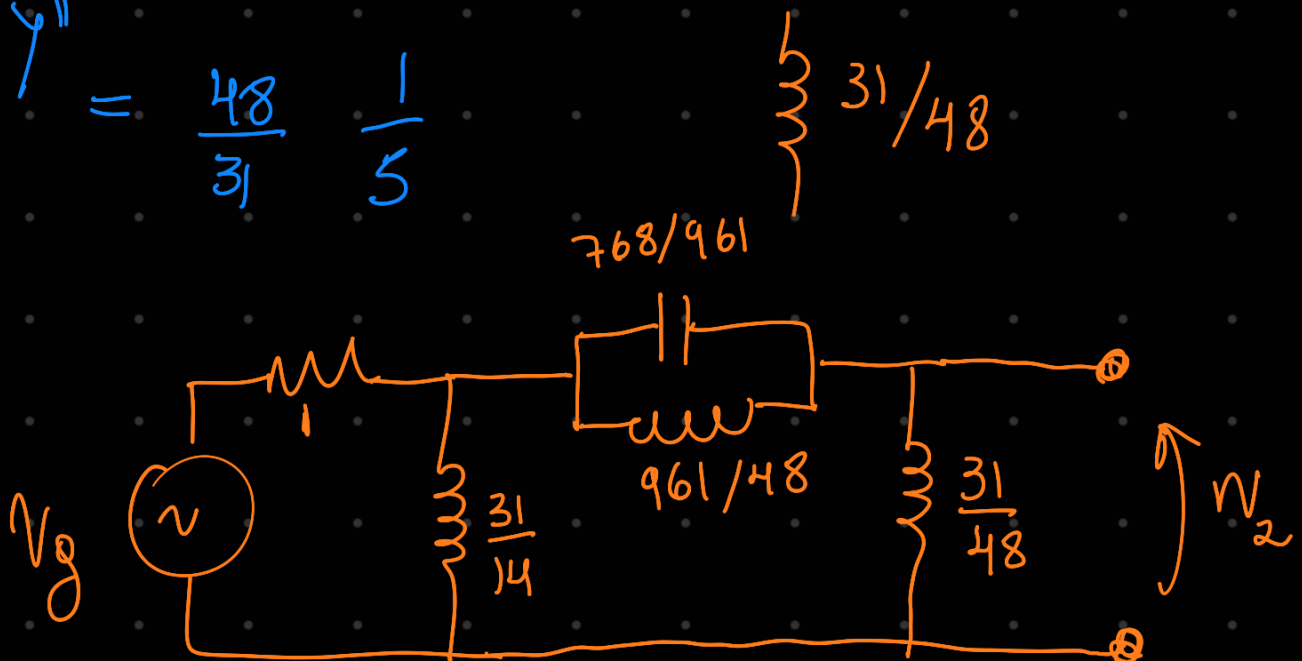
$$L = 16 \cdot \frac{1}{C} = 16 \frac{961}{768} = \frac{961}{48}$$

$$\frac{s^2 LC + 1}{sL} = \frac{s \frac{1}{C}}{s^2 + \frac{1}{LC}}$$

$$Z'' = \frac{\frac{31}{48} (s^3 + 2s)}{s^2 + 1/16} = \frac{961 s}{768}$$

$$= \frac{31}{48} s \left(\cancel{s^2 + \frac{48}{768} 16} \right) = \frac{31}{48}$$

$$Y'' = \frac{48}{31} \frac{1}{s}$$



Verificación en cascada

$$T_+ = \underbrace{\begin{bmatrix} 1 & R_g \\ 0 & 1 \end{bmatrix}}_{\text{resistor}} \underbrace{\begin{bmatrix} 1 & 0 \\ \frac{1}{sL_1} & 1 \end{bmatrix}}_{\text{inductor}} \underbrace{\begin{bmatrix} 1 & \frac{\frac{s}{C_2}}{s^2 + \frac{1}{L_2 C_2}} \\ 0 & 1 \end{bmatrix}}_{\text{parallel LC circuit}} \underbrace{\begin{bmatrix} 1 & 0 \\ \frac{1}{sL_3} & 1 \end{bmatrix}}_{\text{inductor}}$$

$$T_+ = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \rightarrow \begin{cases} V_1 = A V_2 + (-I_2)B \\ I_1 = C V_2 + (-I_2)D \end{cases}$$

$$A = \left. \frac{V_1}{V_2} \right|_{-I_2=0} \quad \therefore \quad \left. \frac{V_2}{V_g} = \frac{1}{A} \right|$$

$$\underbrace{\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}} \begin{bmatrix} 1 & 0 \\ y_2 & 1 \end{bmatrix} \begin{bmatrix} 1 & z_+ \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ y_3 & 1 \end{bmatrix}$$

$$\underbrace{\begin{bmatrix} 1+y_2 & 1 \\ 1 & 1 \end{bmatrix}} \begin{bmatrix} 1 & z_+ \\ 0 & 1 \end{bmatrix}$$

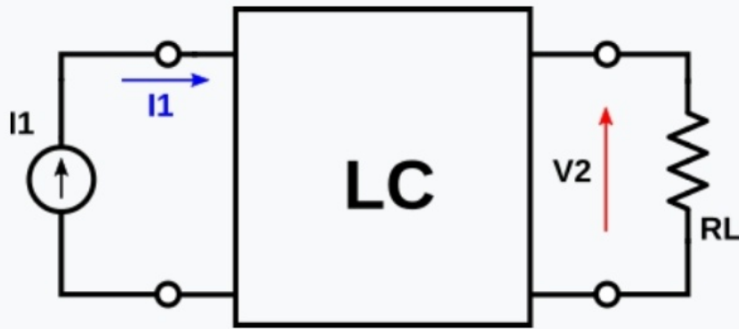
$$\underbrace{\begin{bmatrix} 1+y_2 & (1+y_2)z_+ + 1 \\ 1 & 1 \end{bmatrix}} \begin{bmatrix} 1 & 0 \\ y_3 & 1 \end{bmatrix} \left\{ (1+y_2) + y_3[(1+y_2)z_+ + 1] \right\}$$

$$A = (1 + \gamma_2) + \gamma_3 [(1 + \gamma_2) z_T + 1]$$

→ Verifico con Python $k=0$

$$A = \frac{16s^3 + 5}{16s^3 + 32s^2 + 32s + 16} = \frac{\cancel{16}s(s^2 + \frac{1}{16})}{\cancel{16}[s^3 + 2s^2 + 2s + 1]}$$

2) Dada la siguiente **transferencia de tensión-corriente**:



$$T(s) = \frac{V_2}{I_1} = \frac{k \cdot (s^2 + 9)}{s^3 + 2s^2 + 2s + 1}$$

a) Sintetizar el cuadripolo pasivo sin pérdidas.

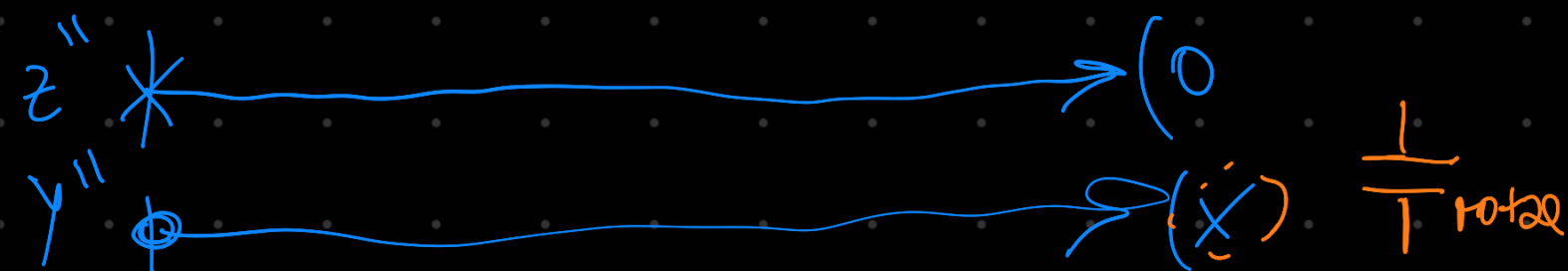
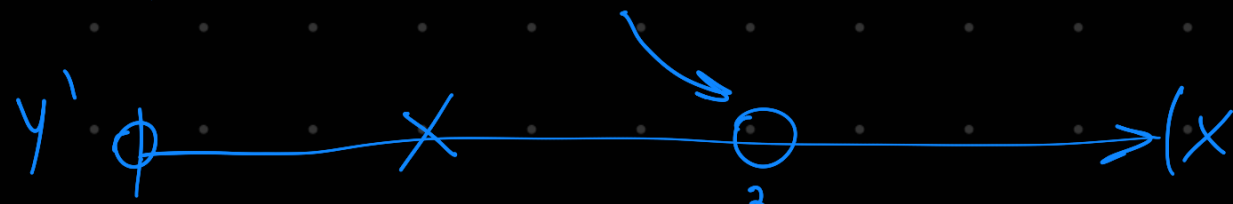
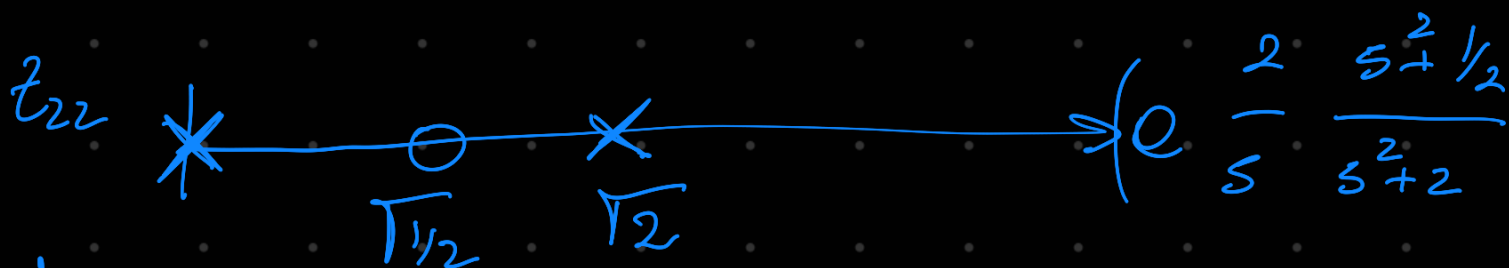
b) Hallar el valor de k que cumple con la síntesis y valor de los componentes hallados.

$$Z = \begin{cases} V_1 = z_{11}I_1 + z_{12}I_2 \\ V_2 = z_{21}I_1 + z_{22}I_2 \end{cases} \quad \begin{aligned} V_2 &= z_{21}I_1 + z_{22} \frac{-V_2}{R_L} \\ V_2 \left(1 + \frac{z_{22}}{R_L}\right) &= z_{21}I_1 \end{aligned}$$

$$\frac{V_2}{I_1} = \frac{z_{21}}{\left(1 + \frac{z_{22}}{R_L}\right)} = \frac{s^2 + 9}{s^3 + 2s^2 + 2s + 1} \quad \text{Par}$$

$$\frac{P}{Q} = \frac{s^2 + 9}{N_a + N_b} = \frac{s^2 + 9}{(2s^2 + 1) + (s^3 + 2s)}$$

$$\frac{V_2}{I_1} = \frac{z_{21}}{1 + z_{22}} = \frac{s^2 + 9}{s^3 + 2s} = \frac{s^2 + 9}{1 + \frac{2s^2 + 1}{s^3 + 2s}} \quad \left. \begin{aligned} & \\ & \end{aligned} \right\} z_{22} \text{ (sintetizar)}$$



$$\varphi' = \frac{1}{z_{22}} - \frac{1}{z_{22}} \Big|_{s_2} = -\varphi$$

$$\frac{s^3 + 2s}{2s^2 + 1} = \frac{s(-9+2)}{-18+1}$$

$$= \frac{s^3 + 2s}{2s^2 + 1} - \frac{7}{17}s \quad \frac{1}{7} \frac{7}{17}$$

$$\frac{s^3 + 2s - \frac{7}{17}s(2s^2 + 1)}{2s^2 + 1} = \frac{\frac{3}{17}s^3 + \frac{27}{17}s}{2s^2 + 1}$$

$$y' = \frac{\frac{3}{17}s(s^2 + 9)}{2s^2 + 1} \rightarrow z' = \frac{17}{3} \frac{2s^2 + 1}{s(s^2 + 9)}$$

$$z'' = z' - \frac{s}{s^2 + 9} \quad / \quad k_{-3} = \frac{17}{3} \frac{2s^2 + 1}{s^2} \Big|_{s^2 = -9}$$

$$\frac{17}{3} \frac{-18 + 1}{-9} = \frac{17^2}{3^3}$$

$$z'' = \frac{17}{3} \frac{2s^2 + 1}{s(s^2 + 9)} - \frac{s^2 \cdot \frac{17^2}{3^3}}{s(s^2 + 9)}$$

$$k_{-3} = \frac{289}{27} \quad -11 \quad 27/289$$

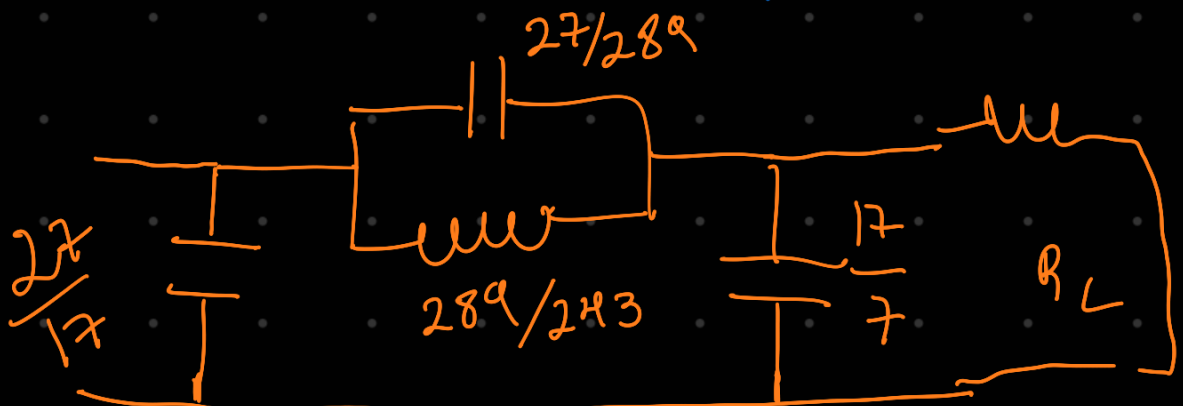
$$= \frac{\left(\frac{17}{27} s^2 + \frac{17}{3} \right)}{s(s^2 + 9)}$$

$$LC = \frac{1}{9}$$

$$L = \frac{1}{9} \cdot \frac{1}{C} = \frac{289}{243} \text{ mm}$$

$$= \frac{\frac{17}{27} (s^2 + 9)}{s(s^2 + 9)}$$

$$Z'' = \frac{17}{27} \frac{1}{s} \rightarrow Y'' = \frac{27}{17} s$$



Verificación en cascada

$$T_+ = \begin{bmatrix} 1 & 0 \\ sC_1 & 1 \end{bmatrix} \begin{bmatrix} 1 & \frac{sC_2}{s^2 + \frac{1}{L_2 C_2}} \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ sC_3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \frac{1}{R_L} & 1 \end{bmatrix}$$

$$T_+ = \begin{bmatrix} A & B \\ C & D \end{bmatrix} \rightarrow \begin{cases} V_1 = A V_2 + (-I_2) B \\ I_1 = C V_2 + (-I_2) D \end{cases}$$

$$\left. \frac{V_2}{I_1} \right|_{-I_2=0} = \frac{1}{C}$$

$$\begin{bmatrix} 1 & 0 \\ \gamma_1 & 1 \end{bmatrix} \begin{bmatrix} 1 & z_2 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \gamma_3 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 1 & 1 \end{bmatrix}$$

$$\begin{bmatrix} \gamma_1 & \gamma_1 z_2 + 1 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ \gamma_3 & 1 \end{bmatrix} = \begin{bmatrix} \gamma_1 + \gamma_3 [\gamma_1 z_2 + 1] & \gamma_1 z_2 + 1 \end{bmatrix}$$

$$C = \gamma_1 + \gamma_3 [\gamma_1 z_2 + 1] + \gamma_1 z_2 + 1$$

